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# Leveraging Local LLMs for Automated Changelog Generation

Christian Goll <cgoll@suse.com>

SUSE

# Quickstart

## Tumbelweed

```
#> sudo zypper in python-changelog-gen
#> cd SUPER_DUPER_PKG
#> osc-ai-vc
```

## Leap 16.0

```
#> sudo zypper ar obs://science:machinelearning sc:ml
#> sudo zypper in python-changelog-gen
#> cd SUPER_DUPER_PKG
#> osc-ai-vc
```

## Resources

https://huggingface.co/mslacken/t5-finetune-changelog
https://github.com/openSUSE/changelog-gen

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# Motivation

## Changelogs in Open Build Service

- Changelog of packages are created from `.changes` file
- Mostly reflect upstream changelog
- **But** must also contain changes for package creation

## The Opportunity

- Accepted requests to `openSUSE:Factory` should all be in the right format
- submission requests contain the necessary information
- Perfect candidate for LLM automation

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# Why Fine-tune a Local LLM?

## Existing tools

- Existing tools can the same
- Can't pickup changes to `.spec` or `_multibuild` files

## Advantages

- **Hype Train**
- pickup semantic changes like avoid to mention changes regarding Apple
- possible to run complete locally, can handle embargoed changes

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# Which model to choose

## Constraints

- model must be small (arround 1Gb)
- permissive license
- not quantized, so that it can be finetuned
- transformer based and no GPT model

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# Encoder-Decoder vs Decoder-Only

## Encoder-Decoder: T5, BART, mT5

- **Bidirectional encoder**: Sees entire input at once
- Best for: Translation, summarization, structured text generation

## Decoder-Only: GPT-3, GPT-4, LLaMA, Mistral

- **Single decoder**: Input and output in same stream but only sees previous tokens
- Best for: Text completion, general-purpose chat

## Key Difference

T5 can **read all input first**, then generate. GPT processes input and output sequentially. This makes T5 more efficient for tasks with structured input.

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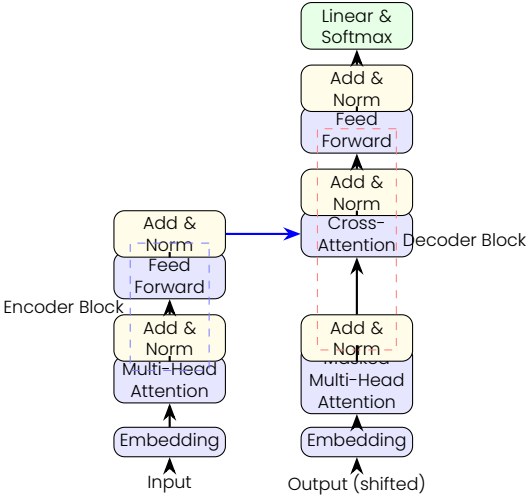
# Encoder-Decoder Architecture (T5)

- **Encoder**: Bidirectional processing of the input sequence.
- **Decoder**: Autoregressive generation with cross-attention to encoder.

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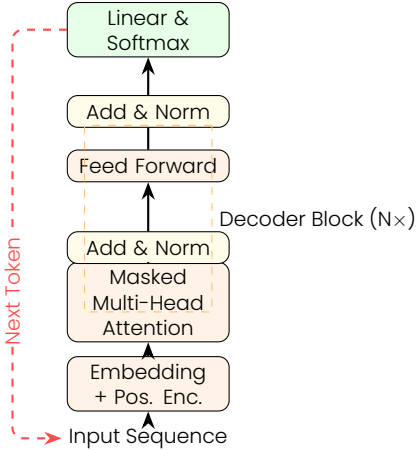
# Decoder-Only Architecture (GPT/LLaMA)

- **Streamlined**: Unified architecture for both input and output.
- **Causal Masking**: Each token only attends to previous tokens.

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# T5: Origins and History

## The T5 Paper (2019) by Google

- **T5 = Text-To-Text Transfer Transformer**
- Systematic study of NLP transfer learning
- **Everything is text-to-text**
- Apache 2.0 License

## Pre-training

- Trained on C4 dataset (Colossal Clean Crawled Corpus)
- 750 GB of cleaned web text

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# T5 Architecture: Encoder-Decoder

## Encoder-Decoder Design

- **Encoder**: Processes input text
  - Bidirectional attention
  - Understands full context
  - Creates rich representations
- **Decoder**: Generates output text
  - Autoregressive generation
  - Attends to encoder outputs
  - Produces token-by-token

## Model Sizes

| Model | Params | Layers | Size |
|---|---|---|---|
| Small | 60M | 12 | 242 MB |
| Base | 220M | 24 | 892 MB |
| Large | 770M | 48 | 2.75 GB |

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# OBS Data Extraction

## What data do we need?

- Accepted requests to `openSUSE:Factory`
- Spec file diffs (version updates, build changes)
- Source file changes (added, removed, modified)
- Release notes from upstream (GitHub, archive changelogs)
- The actual `.changes` file entries (our target)

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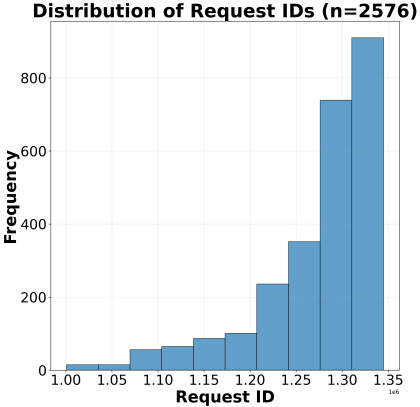
# OBS Data Extraction

## Naive approach

- Choose a random number smaller than latest request
- Check if submission went to `openSUSE:Factory` and was accepted
- Identify `Source0`
- Extract first 20 lines of `CHANGELOG` …
- For github links fetch github release page




use half side gaussian to get recent requests

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# OBS Data Extraction

## Privacy considerations

- Spec file is MIT license and content can be used
- Changes file is also MIT license
- `CHANGELOG` files fall under OSS license if accepted to `openSUSE:Factory`
- github release page is the same licese as the source

## Personal information

- Submitter name and email is removed from changes files
- Reviewer isn't retrieved
- Request is used as key but not used for training

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# OBS Data Extraction - Example Update


```
{ "request_id": 1111873, "project": "GNOME:Factory", "package": "gnome-music", "version": "45.0",
   "changed_files": [ "gnome-music-45.0.tar.xz", "gnome-music.changes", "gnome-music.spec" ],
   "removed_files": [],
   "added_files": [],
   "spec_diff": "Name:            gnome-music\n-Version:        44.0\n+Version:        45.0\n Release:
    0\n Summary:        Music Player for GNOME\n License:        LGPL-2.1-or-later AND SUSE-GPL-2.0-with-
    plugin-exception\n Group:            Productivity/Multimedia/Sound/Players\n URL:            https://www.
    gnome.org\n-Source0:        https://download.gnome.org/sources/gnome-music/44/%{name}-%{version}.tar.xz\n
    +Source0:        https://download.gnome.org/sources/gnome-music/45/%{name}-%{version}.tar.xz\n #PATCH-FIX
    -SLE  0001-gnome-music-use-python36.patch yfjiang@suse.com -- disable python 3.7 specific feature to
    allow gnome-music build and run on python 3.6 for SLE/Leap 15.4.\n Patch0:         0001-gnome-music-use-
    python36.patch",
   "changes_diff": "+\n+- Update to version 45.0:\n+  + Minor quality of life improvements.\n+  + Updated
    translations.\n+\n - Update to version 44.0:",
   "source_file": "gnome-music-45.0.tar.xz",
   "archive_changelog": "Overview of changes in 45.0\n==========================\n\•n Minor quality of life
    improvements\n\nThanks to our contributors this release:\n Carlos Garnacho\n Jean Felder\n Harold
    Knowlden\n Marinus Schraal\n\nUpdated or added translations:\n Abkhazian\n Romanian\n Italian\n Turkish\n
     British English\n French\n Hungarian\n German\n Hebrew\n Ukrainian",
   "url": "https://www.gnome.org",
   "github_release_notes": "" },
```


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# OBS Data Extraction - Example Update Rendered

```
<INPUT>
create structured changelog for package gnome-music from 44.0 to 45.0:
changelog:
Overview of changes in 45.0
 ==========================
 • Minor quality of life improvements
 Thanks to our contributors this release:
 Carlos Garnacho
 Jean Felder
 Harold Knowlden
 Marinus Schraal
 Updated or added translations:
 Abkhazian
 [....]
 Hebrew
 Ukrainian
changes in spec file:
-Source0: https://download.gnome.org/sources/gnome-music/44/%<unk>name<unk>-%<unk>version<unk>.tar.xz
 +Source0: https://download.gnome.org/sources/gnome-music/45/%<unk>name<unk>-%<unk>version<unk>.tar.xz
</INPUT>
--------------------------------------------------------------------------------
<TARGET>
- Update to version 45.0:
  + Minor quality of life improvements.
  + Updated translations.
</TARGET>
```

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# OBS Data Extraction - Example Patch File


```
{ "request_id": 1071207, "project": "network:utilities", "package": "pdsh", "version": "2.34",
  "changed_files": [ "pdsh.changes", "pdsh.spec" ],
  "removed_files": [],
  "added_files": [ "Add-call-to-slurm\_init-this-makes-sure-the-config-options-are-set.patch" ],
  "spec_diff": "Patch6:         fail-fast-on-ssh-errors-or-non-zero-return-code.patch\n Patch7:         doc-
   fast-fail-update.patch\n Patch8:         Hack-to-work-around-a-generic-type-name-breakage-introduced-by-
   latest-Slurm.patch\n+Patch9:         Add-call-to-slurm\_init-this-makes-sure-the-config-options-are-set.
   patch\n %description\n Pdsh is a multithreaded remote shell client which executes commands on\n %patch6 -
   p1\n %patch7 -p1\n %patch8 -p1\n+%patch9 -p1\n %build\n export CFLAGS=\"%{optflags} -fno-strict-aliasing
   \"",
  "changes_diff": "+\n+- Fix slurm plugin: make sure slurm\_init() is called before using\n+  the Slurm API.
    This has been the case since version 20.11 (bsc#1209216).\n+  Add-call-to-slurm\_init-this-makes-sure-
   the-config-options-are-set.patch\n+\n - Hack-to-work-around-a-generic-type-name-breakage-introduced-by-
   latest-Slurm.patch",
  "source_file": "pdsh-2.34.tar.gz",
  "archive_changelog": "This file describes changes in recent versions of pdsh. It primarily\ndocuments
   those changes that are of interest to users and admins.\n\n* Changes in pdsh-2.34 (2020-01-07)\n
   ==================================\n -- Fix for output corruption with no newlines (#114)\n -- pipecmd:
   fix result check error handling (Dylan Simon)\n -- slurm: workaround slurm export of internal List
   interfaces\n -- readline: add application name as argv[0] (#112)\n -- Fix errors from lgtm.com scan\n\n*
   Changes in pdsh-2.33 (2017-06-28)\n==================================\n -- Fix segfault and build issues
    on Mac OSX (#95)\n -- Always pass RTLD_GLOBAL to dlopen(3) of modules. Fixes missing symbol\n     errors
    from modules using libraries that also use dlopen() (e.g.\n     nodeupdown, slurm)\n\n* Changes in pdsh
   -2.32 (2017-06-22)\n==================================\n -- Autotools update\n -- Switch to dlopen(3)/
   dlsym(3) instead of using libltdl",
  "url": "http://pdsh.googlecode.com/",
  "github_release_notes": ""
},
```


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# OBS Data Extraction - Example Patch File Rendered

```
<INPUT>
create structured changelog for package pdsh 2.34:
new files: ['Add-call-to-slurm\_init-this-makes-sure-the-config-options-are-set.patch']
changes in spec file:
+Patch9: Add-call-to-slurm\_init-this-makes-sure-the-config-options-are-set.patch
 +%patch9 -p1
</INPUT>
--------------------------------------------------------------------------------
<TARGET>
- Fix slurm plugin: make sure slurm\_init() is called before using
  the Slurm API. This has been the case since version 20.11 (bsc#1209216).
  Add-call-to-slurm\_init-this-makes-sure-the-config-options-are-set.patch
</TARGET>
```

## Note

Upstream CHANGELOG missing on purpose, requests adds patch file

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# Data Preprocessing & Tokenization

## Special Token Handling

- Newline character (`\n`) added as special token
- Preserves structure in multi-line changelogs

## Sequence Length Limits

- **Input**: Maximum 1024 tokens (spec diffs + changelog)
- **Target**: Maximum 512 tokens (generated changelog entry)
- Sequences exceeding limits are truncated

## Padding Strategy

- Dynamic padding: sequences padded to `longest` in batch
- Padding tokens in labels replaced with $-100$
- Loss function ignores $-100$ (no gradient from padding)

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# Untrained Models: Fine Tuning is necessary

1071207

- **small**: `:-pdsh 2.34::-call-to-slurm_init-this-makes-sure-the-config-options-a aeoe o аи ереади в ил и рораии ирадиии орамителноо`
- **base**: `structured changelog for package pdsh 2.34:increasing structured changelogging for package create structured change log for package build pcsh pkg 2.34.34:Auffinden Sie neue files: ['Add-call-to-slurm_init-this-makes-sure-the-config-options-ar +%patch9 -p1: Add-call/slursm_install-this`
- **large**: `-p1`

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